Closed-Form 6-DoF Pose Estimation of Medical Tools via Gradiometer-Based Electromagnetic Localization System (Supplementary)

Zhiwei Wu, Jiahao Luo, Yubo Pu, Siyi Wei, Yuankai Chen, and Jinhui Zhang

I. Source Geometry Metric and Manipulator Reconfiguration

A. Simulation Construction and Error Statistics

Using $g = g_{\mathrm{shape}}$ defined in (15) of the main manuscript, we generate three centered source positions in the xy plane as

$$\boldsymbol{P}_C(g) = \frac{s_C}{\sqrt{1+g^2}} \begin{bmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ g/\sqrt{6} & -2g/\sqrt{6} & g/\sqrt{6} \\ 0 & 0 & 0 \end{bmatrix}.$$

Here $s_C = 0.33\,\mathrm{m}$ fixes $\|\boldsymbol{P}_C\|_F$ and the root-mean-square distance $s_C/\sqrt{3}$ from the zero centroid. Decreasing g from 1 to 0.01 contracts the y direction and expands the x direction, flattening the equilateral triangle toward collinearity. Individual source-to-target distances are not fixed.

The continuous sweep evaluates 25 geometries using the hardware positions of the 12 QMC6309 sensors and three first-order dipoles of $250\ \mathrm{A\,m^2}$ each. The same 100 target positions are sampled by a deterministic Latin hypercube in an $80 \times 80 \times 80\ \mathrm{mm}^3$ workspace centered at $(0, 0, -300)\ \mathrm{mm}$, with independently sampled orientations. Each target is evaluated in 30 trials with independent, zero-mean Gaussian perturbations of $0.5\,\mu\mathrm{T}$ standard deviation per sensor axis and electromagnet measurement. Target poses and perturbation sequences are reused across geometries for paired comparisons.

Errors $e_{\boldsymbol{p}}$ and $e_{\boldsymbol{R}}$ are first averaged over the 30 trials for each target. The plotted mean and 10th–90th percentile bands are then computed across targets. The bands describe variation across target poses, not confidence intervals.

B. Localization Evaluation at Representative Geometries

The reconfiguration objective and joint update are given in (16) of the main manuscript. For the five representative geometries in Fig. 1, collision-checked MoveIt plans realize triangular frustum layouts. Each state is evaluated on the same 200 target poses in a $240 \times 240 \times 80\ \mathrm{mm}^3$ workspace, using the same magnetic model, noise level, and 30 trials per pose as above. Fig. 1 shows the configurations and position errors, and Tab. I summarizes the error statistics. Independent reruns reproduced both studies exactly.

II. Source Distance and Gradient Resolvability

A. Controlled Distance Simulation

We simulate point dipoles using the source positions and tracked magnetic axes averaged over the Fig. 6 poses and the eight-sensor Config. 2 geometry. From a recorded pose, the array moves away along the source-plane normal with fixed orientation, increasing all three source distances over 41 positions (mean distance: $205.9\,\mathrm{mm}$ to $800\,\mathrm{mm}$). Two fixed moment settings are used: $94.684\ \mathrm{A\,m^2}$ per source, giving a mean field magnitude of $2\,\mathrm{mT}$ over the reference poses, and 53.748, 46.685, and $65.243\ \mathrm{A\,m^2}$ fitted to field vectors from five chronological records.

TABLE I
Localization error across the five representative source geometries.

g_{shape}	Mean $e_{\boldsymbol{p}}$ [mm]	Median [mm]	P10–P90 [mm]	Mean $e_{\boldsymbol{R}}$ [deg]
1.00	11.76	11.06	6.24–18.10	2.93
0.50	13.96	12.90	7.54–22.20	3.41
0.20	26.85	23.68	13.55–44.00	5.78
0.05	109.11	99.62	51.26–186.72	22.43
0.01	290.56	289.04	218.60–368.13	68.11

Each setting uses 200 realizations of static CV measurements with three excitation slots and one shared background slot. Each realization is solved using one measurement per sensor per time slot. We assume independent Gaussian noise of $0.3\,\mu\mathrm{T}$ standard deviation per sensor axis in each slot. After background subtraction, each source receives an effective residual field discrepancy with zero spatial mean and a vector RMS of $6\,\mu\mathrm{T}$ across sensors. Noise and residual draws are reused across distances and moment settings. We quantify gradient signal quality by the signal to total reconstruction error ratio

$$20\log_{10}\frac{\sqrt{\sum_{k=1}^{3}\|\boldsymbol{X}_k\|_F^2}}{\sqrt{\frac{1}{200}\sum_{t=1}^{200}\sum_{k=1}^{3}\|\hat{\boldsymbol{X}}_{k,t}-\boldsymbol{X}_k\|_F^2}}. \tag{1}$$

Here $\boldsymbol{X}_k$ is the exact dipole gradient at the array center and $\hat{\boldsymbol{X}}_{k,t}$ is its estimate in realization t. The total error includes slot noise, residual discrepancy, and the finite-array approximation. For random noise SNR, the denominator is replaced by the RMS gradient error from slot noise alone.

Both moment settings show increasing position error and decreasing gradient signal quality with distance (Fig. 2). For the $2\,\mathrm{mT}$ reference, increasing the mean distance from $205.9\,\mathrm{mm}$ to $399.0\,\mathrm{mm}$ raises the median position error from $10.0\,\mathrm{mm}$ to $354.4\,\mathrm{mm}$ and lowers the gradient signal to total error ratio from $23.3\,\mathrm{dB}$ to $-1.5\,\mathrm{dB}$. At $800\,\mathrm{mm}$, the random noise SNR falls to $-7.7\,\mathrm{dB}$. These results show how gradient attenuation relative to the prescribed measurement errors amplifies position error.

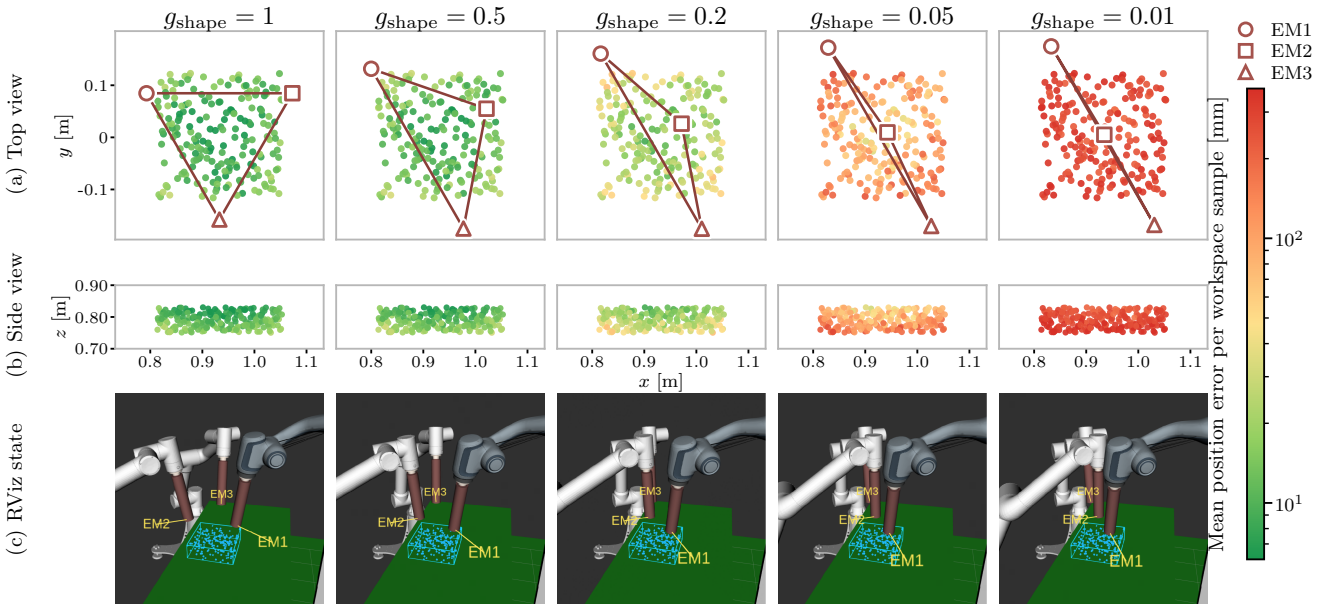


Fig. 1. Five representative source geometries and localization results. The columns correspond to $g_{\text{shape}} = 1$, 0.5, 0.2, 0.05, and 0.01. From top to bottom, the rows show (a) top views of the source positions and target samples colored by mean position error, (b) side views over the displayed 0.70–0.90 m height range, and (c) the corresponding RViz configurations. Circle, square, and triangle markers distinguish the three electromagnets.

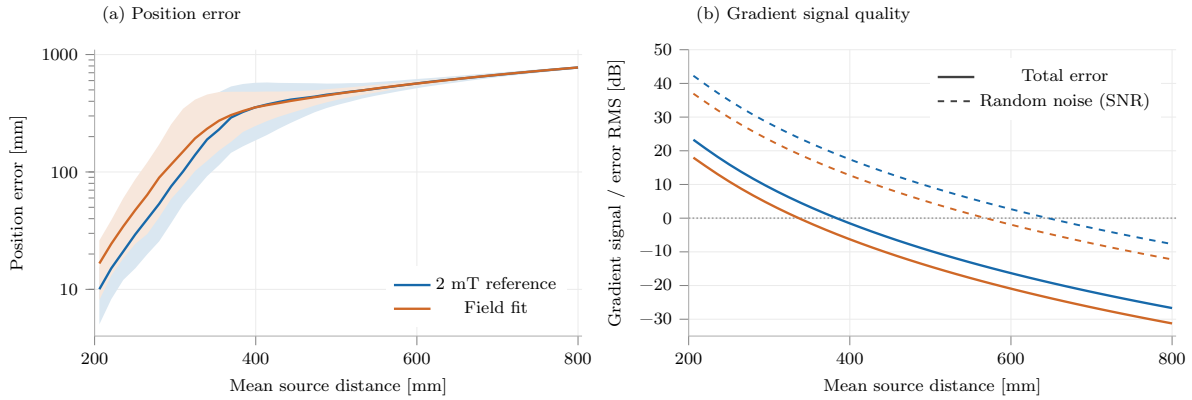


Fig. 2. Controlled distance simulation. Colors distinguish the 2 mT reference and field-fit moment settings. (a) Median position error with 10th–90th percentile bands over 200 realizations. (b) Gradient signal to total error ratio (solid) and random noise SNR (dashed). The horizontal axis is the mean of the three source-to-array distances.

III. Motion Artifacts under Sequential Sampling

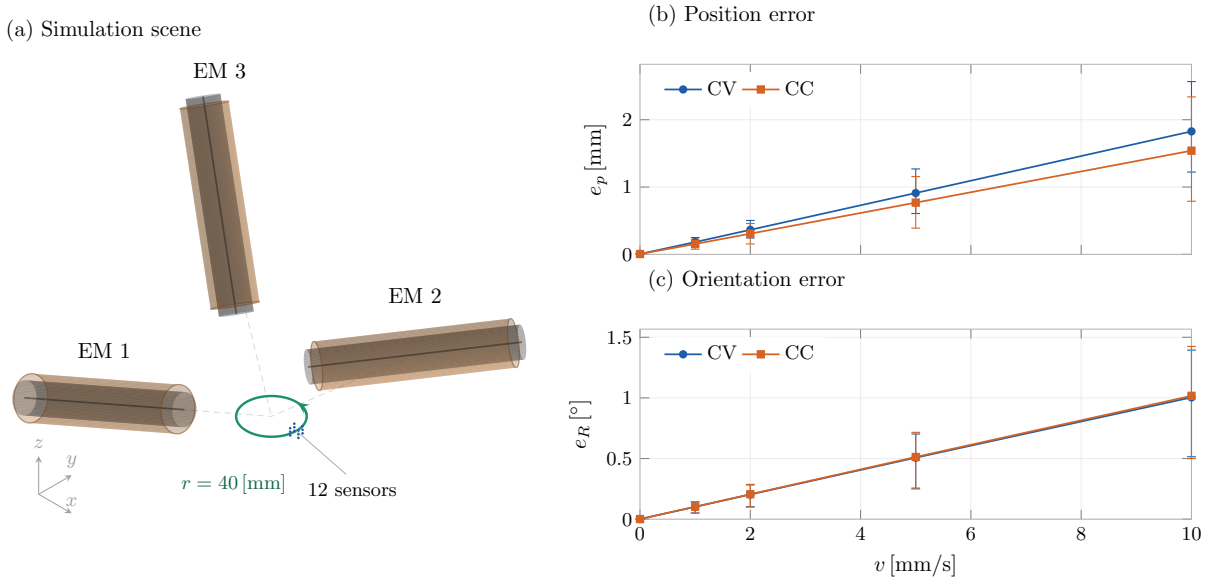


Fig. 3. Noiseless point-dipole simulation with the 12-sensor configuration and source geometry from Fig. 6. (a) Circular translation of radius 40 mm with fixed array orientation. Sensor spacing is enlarged for visibility. (b) Position error and (c) orientation error for CV and CC sampling. Markers show means and error bars span the 5th–95th percentiles over 180 matched trajectory phases at each speed. The nominal CV/CC slot durations are 72/32 ms, with cycle periods of 288/96 ms. Each sensor is sampled at the midpoint of its successive 1 ms window. Errors are referenced to the end of the last active-source acquisition window.

TABLE II

Qualitative requirements and reported performance of representative magnetic localization methods.

Reference	DoF	Val.	Map-free[†]	Init.-free	Iter.-free	Source requirement	Workspace/ envelope	e_p (mm)	e_R (°)	t_{solve} (ms)	f_{sys} (Hz)
Fischer [1]♣	2	Exp.	✗	✗	✗	8 EM	25-mm local	2.74–7.94	–	38.3^a	10^a
Fischer [2]♠	6	Sim.	✗	✗	✗	8 EM	20–30-mm local	3.15 ± 1.62	3.67 ± 3.65	NR	NR
Di Natali [3]♣	6	Exp.	✗	✗	✗	1 PM	150-mm cyl.	< 7	≤ 5	1^b	$> 100^b$
von Arx [4]♢	6	Exp.	✗	△	✗	3 EM	$80 \times 80 \times 60$ mm	0.27–0.58	0.48–1.83	NR	≤ 10
Xiang [5]♢	6	Exp.	✗	✓	✗	≥ 4 poses	NR	$0.12/0.17^c$	$0.18/0.26^c$	NR	NR
GELS♡	6	Exp.	✓	✓	✓	≥ 3 EM	$274 \times 175 \times 10$ mm	3.07^d	3.79^d	0.172	3.47–14.49

Quadrant: ♠ static field + gradiometer; ♡ dynamic field + gradiometer; ♣ static field + magnetometer; ♢ dynamic field + magnetometer. For a multistage method, the symbol denotes the stage compared in the table. ✓: property satisfied; ✗: property not satisfied; △: not clearly specified; **NR**: not reported. [†]**Map-free**: no pre-calibrated workspace field map or identified source field model or parameters. **Init.-free**: no externally supplied initial pose estimate is required; an internally generated initializer is allowed. **Iter.-free**: no online iterative pose optimization. **Source requirement**: the physical source count and, where needed, the source poses or motion used by the localization procedure. [a]Position-estimation time and sensor sampling rate; computing a new field and gradient setting required 579 ms. [b]The localization algorithm required 1 ms, while the full acquisition and communication cycle required 7 ms. [c]The paper reports no stand-alone NSS initialization error. The listed values are post-L-M patch and internal-magnet errors, respectively, and the latter uses sensor poses returned by the first-stage L-M optimization. [d]Independently best sequence-level GELS values in Fig. 6; they occurred in different Config. 2 CC sequences. Across the six configuration/mode settings, the sequence-level means were 10.80–15.57 mm and 11.5°–19.9°.

References

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